

Sheth L.U.J and Sir M.V. college

Loops in python

Assignment

10. Guess the number game: Create a simple game where the user has to guess a randomly generated number. The program should provide hints ("too high." "too low") and continue looping until the user guesses correctly.

For loops:

- 1) Sum of N numbers: Write a program that uses a for loop to calculate the sum of the first 100 positive integers.

```
#Sum of N numbers: Write a program that uses a for loop to calculate the sum of the first 100 positive integer
limit = int(input("Enter the limit to find sum "))
sum=0
for i in range(1,limit+1):
    sum = sum + i
print("Sum of numbers = ",sum)
```

```
Enter the limit to find sum 10
Sum of numbers = 55
```

- 2) Factorial: Calculate the factorial of a given number n using a for loop.

```
5 ▶ #Factorial: Calculate the factorial of a given number n using a for loop.
# Example : factorial of 4 = 1*2*3*4
number = int(input("Enter the number to find factorial "))
factorial=1
for i in range(1,number+1):
    factorial = factorial * i
print("Factorial of number ", number, " = ",factorial)
```

```
... Enter the number to find factorial 5
Factorial of number 5 = 120
```

- 3) Count vowels: Write a program to count the number of vowels (a, e, i, o, u) in a given string.

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```
string = input("Enter Text to count vowel : ")
string = string.lower()

count = 0
for s in string :
    if (s=='a' or s=='e' or s=='i' or s=='o' or s=='u'):
        count=count+1
print("Number of vowels in string = ",count)
```

```
Enter Text to count vowel : Tasmiya
Number of vowels in string =  3
```

- 4) List of even numbers: Create a program that iterates through a list of numbers and prints only the even numbers.

```
▶ for z in range(0,20,2):
    print(f"Even numbers:{z}")
```

```
... Even numbers:0
     Even numbers:2
     Even numbers:4
     Even numbers:6
     Even numbers:8
     Even numbers:10
     Even numbers:12
     Even numbers:14
     Even numbers:16
     Even numbers:18
```

- 5) Find the largest number: Find the largest number in a list without using the max() function.

```
#find the largest number: find the largest number in a list without using the max() function.
number=[45,2,6,7,8,9,10]
largest = number[0]
for n in number:
    if n > largest:
        largest = n
print("largest number in the list=",largest)
```

```
largest number in the list= 45
```

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- 6) Power of the number: create a list and input power value from each user. Print the power of each element of the list.

```
#power of a number: create a list and input power vvalue from user. print the power of each element of the list.
numbers=[2,6,8,9,3,4]
pow=int(input("enter the power value="))
for n in numbers:
    print(n,"^",pow,"=",n**pow)
```

```
enter the power value=2
2 ^ 2 = 4
6 ^ 2 = 36
8 ^ 2 = 64
9 ^ 2 = 81
3 ^ 2 = 9
4 ^ 2 = 16
```

While loops:

- 1) Guess the number game: Create a simple game where the user has to guess a randomly generated number. The program should provide hints ("too high," "too low") and continue looping until the user guesses correctly.

```
# Generate a random number between 1 and 100
target_number = random.randint(1, 100)
guess = None
attempts = 0

print("Welcome to the Guess the Number game!")
print("I'm thinking of a number between 1 and 100. Can you guess it?")

while guess != target_number:
    try:
        guess = int(input("Enter your guess: "))
        attempts += 1

        if guess < target_number:
            print("Too low! Try again.")
        elif guess > target_number:
            print("Too high! Try again.")
        else:
            print(f"Congratulations! You guessed the number {target_number} in {attempts} attempts!")
    except ValueError:
        print("Invalid input. Please enter a whole number.")
```

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```
Welcome to the Guess the Number game!
I'm thinking of a number between 1 and 100. Can you guess it?
Enter your guess: 200
Too high! Try again.
Enter your guess: -2
Too low! Try again.
Enter your guess: 56
Too high! Try again.
Enter your guess: 20
Too high! Try again.
Enter your guess: 8
Too low! Try again.
Enter your guess: 10
Congratulations! You guessed the number 10 in 6 attempts!
```

2) Digit sum: Calculate the sum of the digits of a number using a while loop.

```
# sum of the digits of given number
n = int(input("Enter the number : "))
sum = 0
while n>0:
    digit = n%10
    sum +=digit
    n = n//10

print("Sum of the digits = ",sum)
```

```
Enter the number : 456
Sum of the digits = 15
```

3) Palindrome check: Determine if a number is a palindrome (e.g., 121, 343) using a while loop.

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```
# Number Palindrom
n = int(input("Enter the number : "))
org = n
rev = 0
while n>0:
    digit = n%10
    rev = (rev*10) + digit
    n = n//10

print("Reverse of the number = ",rev)
if org == rev:
    print(org," is Palindrome")
else :
    print(org," is not Palindrome")
```

```
Enter the number : 101
Reverse of the number =  101
101 is Palindrome
```

- 4) Fibonacci series: Generate the Fibonacci series up to a given number of terms.

```
num_terms = int(input("Enter the number of terms for Fibonacci"))

n1, n2 = 0, 1
count = 0

# Check if the number of terms is valid
if num_terms <= 0:
    print("Please enter a positive integer.")
elif num_terms == 1:
    print("Fibonacci series up to", num_terms, ":")
    print(n1)
else:
    print("Fibonacci series:")
    while count < num terms:
```

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```
while count < num_terms:
    print(n1, end=" ")
    nth = n1 + n2
    # Update values
    n1 = n2
    n2 = nth
    count += 1
```



```
Enter the number of terms for Fibonacci series: 4
Fibonacci series:
0 1 1 2
```

5) Reverse a number: Reverse a given integer (e.g., 123 becomes 321) using a while loop.

```
# Reverse of given number
n = int(input("Enter the number : "))
rev = 0
while n>0:
    digit = n%10
    rev = (rev*10) + digit
    n = n//10

print("Reverse of the number = ",rev)
```

```
Enter the number : 567
Reverse of the number = 765
```